

到影响。同时通过控制动力的大小而改变飞行器的飞行状态。已知：飞行器的动力 F 始终与飞行方向相同，空气升力 F_1 与飞行方向垂直，大小与速度的平方成正比，即 $F_1 = C_1 v^2$ ；空气阻力 F_2 与飞行方向相反，大小与速度的平方成正比，即 $F_2 = C_2 v^2$ 。其中 C_1 、 C_2 相互影响，可由运动员调节，满足如图 1 所示的关系。飞行员和装备的总质量为 90kg 。（重力加速度取 $g = 10\text{m/s}^2$ 。）

(1) 若飞行员使飞行器以 $v_1 = 10\sqrt{3}\text{m/s}$ 速度在空中沿水平方向匀速飞行，如图 2 所示。则飞行器受到动力 F 大小为多少？

(2) 若飞行员关闭飞行器的动力，使飞行器匀速滑行，且滑行速度 v_2 与地平线的夹角 $\theta = 30^\circ$ ，如图 3 所示，则速度 v_2 的大小为多少？（结果可用根式表示）

(3) 若飞行员使飞行器在空中的某一水平面内做匀速圆周运动，如图 4 所示，在此过程中 C_2 只能在 $1.75 \sim 2.5\text{N}\cdot\text{s}^2/\text{m}^2$ 之间调节，且 C_1 、 C_2 的大小与飞行器的倾斜程度无关。则飞行器绕行一周动力 F 做功的最小值为多少？（结果可保留 π ）

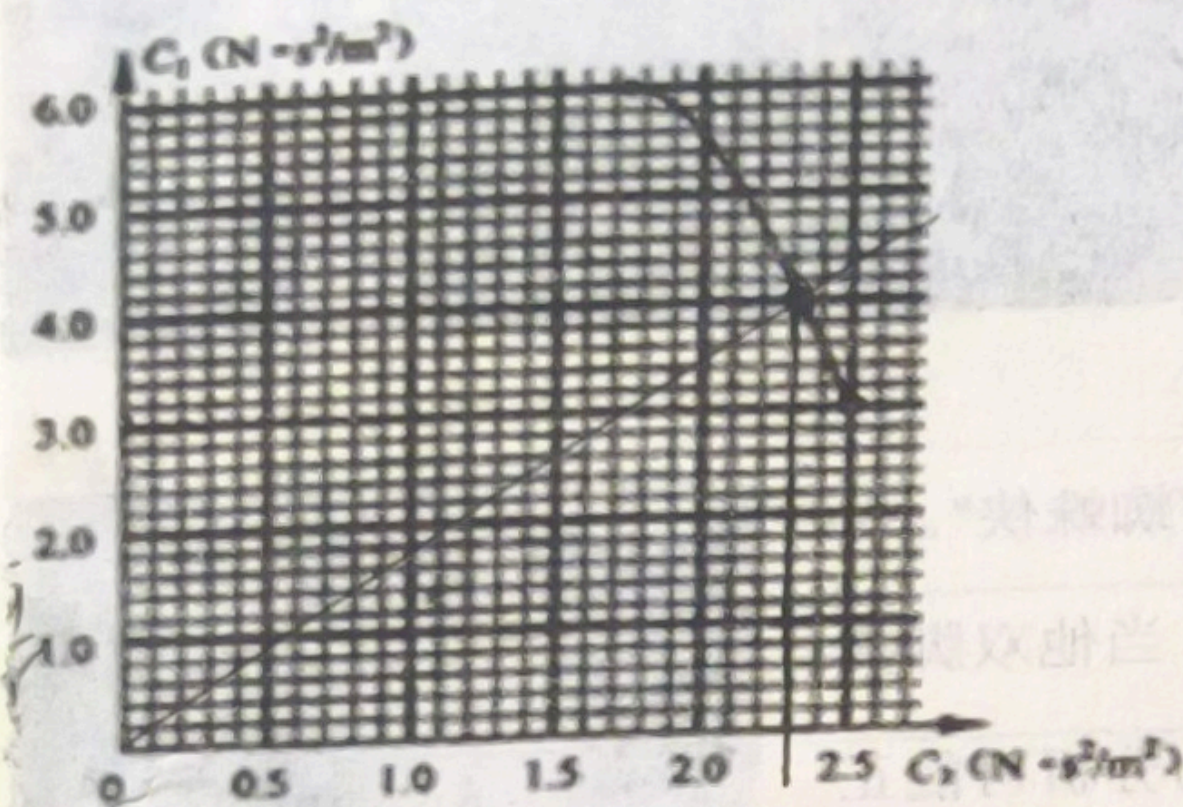


图 1

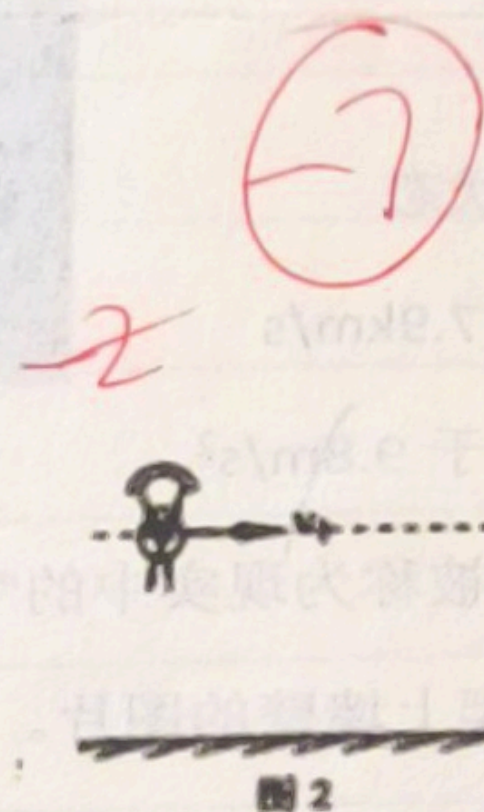


图 2

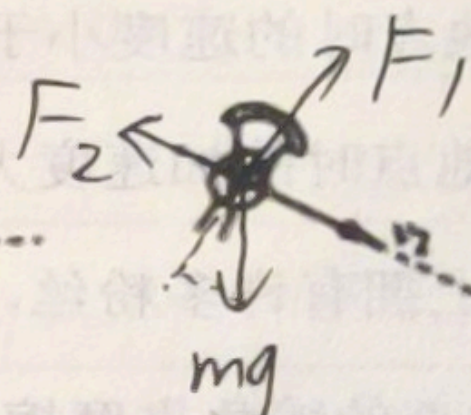


图 3

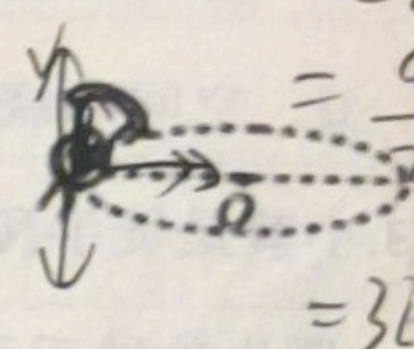


图 4

平衡

$$(1) \quad mg = C_1 v_1^2 \quad C_1 = 3 \text{ N}\cdot\text{s}^2/\text{m}^2$$

$$\therefore C_2 = 2.5 \text{ N}\cdot\text{s}^2/\text{m}^2$$

$$\text{动力 } F = F_2 = C_2 v_1^2 = 750 \text{ N}$$

平衡

$$(2) \quad mg \cos \theta = C_1 v_2^2$$

$$mg \sin \theta = C_2 v_2^2$$

$$C_1 = v_2^2 \cot \theta$$

$$C_2 = 2.3 \text{ N}\cdot\text{s}^2/\text{m}^2 \quad C_1 = 4 \text{ N}\cdot\text{s}^2/\text{m}^2$$

$$v_2 = \sqrt{\frac{225\sqrt{3}}{2}} \text{ m/s}$$

(3) 非平衡

设 F_1 与竖直方向夹角 α

$$mg = C_1 v^2 \cos \alpha$$

$$\text{水平合力提供向心力 } C_1 v^2 \sin \alpha = \frac{mv^2}{R}$$

$$\text{动力 } F = F_2 = C_2 v^2$$

$$W = F \cdot 2\pi R = \frac{2\pi C_2 m^2 g}{C_1^2 \sin \alpha \cos \alpha}$$

$$\alpha = 45^\circ \quad C_1 = 6 \text{ N}\cdot\text{s}^2/\text{m}^2$$

$$C_2 = 1.75 \text{ N}\cdot\text{s}^2/\text{m}^2$$

时 W 最小

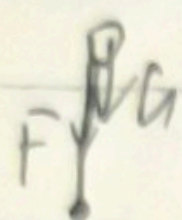
$$W_{\min} = 15750\pi$$

圆周运动 NO.

- 一、确定轨道平面、圆心位置、轨迹圆半径
- 二、受力分析、找向心力、正交分解、指向圆心或背离圆心的力
- 三、列方程求解。
 - 点、半二
 - 过程、动能定理

绳模型

绳只能拉不能推，圆心以上速度一定不为0



最高点 $F + G = \frac{mv^2}{L}$ $F \geq 0 \therefore v \geq \sqrt{gL}$

最低点 $F - G = \frac{mv^2}{L}$ $v^2 = v_0^2 - 4gL \geq gL$

$mg2L = \frac{1}{2}m(v_0^2 - v^2)$ $v_0 > \sqrt{5gL}$

拉力差 $\Delta F = 6mg$

$mg + F_1 = \frac{mv^2}{L}$

$F_2 - mg = \frac{mv^2}{L}$

$2mgL = \frac{1}{2}m(v_2^2 - v_1^2)$

不满足 $v \geq \sqrt{gL}$ 时.

$v_0 = \sqrt{\frac{gL}{2}} < \sqrt{gL}$

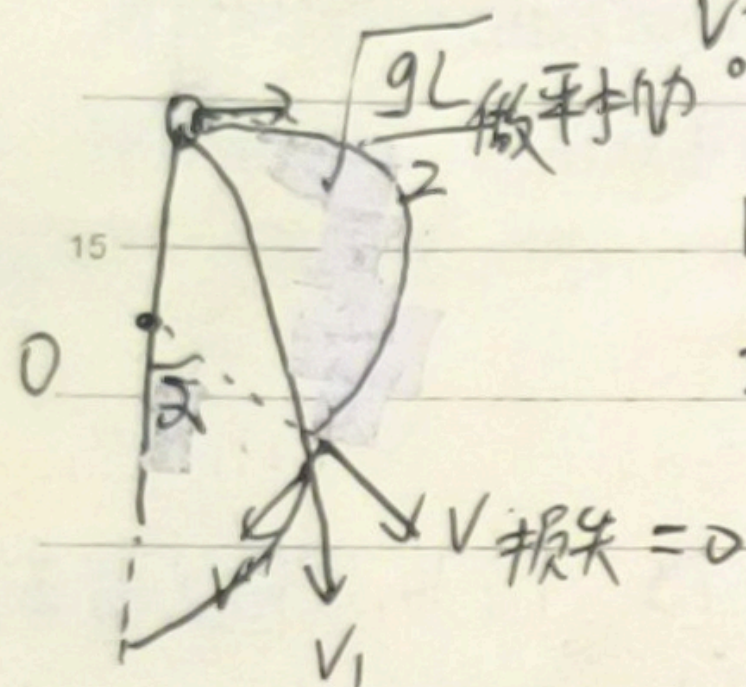
$v_0^2 t^2 = L^2 \sin^2 \alpha$

$\frac{1}{2}gt^2 = L(1 + \cos \alpha)$

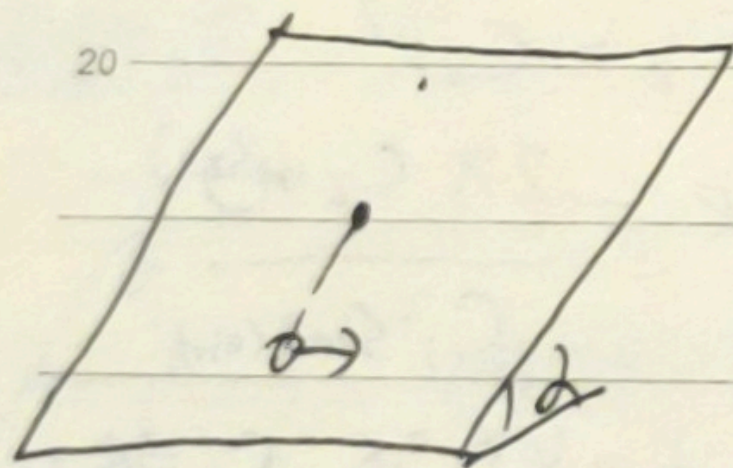
$\Rightarrow d = 90^\circ$ 圆心等高处 绳垂直

$mgL = \frac{1}{2}(v^2 - v_y^2)$ $v^2 = v_y^2 + 2gL = 4gL$

$F - mg = \frac{mv^2}{L} \Rightarrow F = 5mg$



等效重力加速度



$F + mgsin\alpha = \frac{mv^2}{L}$

$F \geq 0$

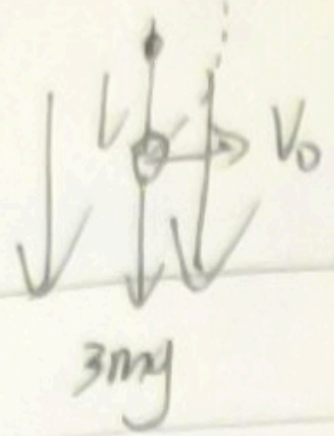
$v_{\perp} \geq \sqrt{gLsin\alpha}$

$v_{\parallel} \geq \sqrt{5gLsin\alpha}$

$mg2Lsin\alpha = \frac{1}{2}m(v_{\parallel}^2 - v_{\perp}^2)$

$v_{\perp}^2 = v_{\parallel}^2 - 4gLsin\alpha > gLsin\alpha$

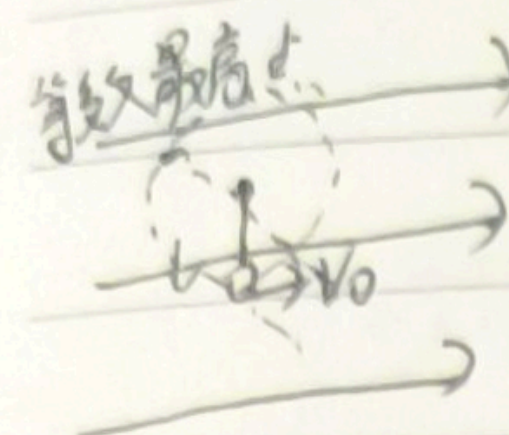
$\Delta F_{\perp\parallel} = 6mgsin\alpha$



$$F = \frac{2mg}{q}$$

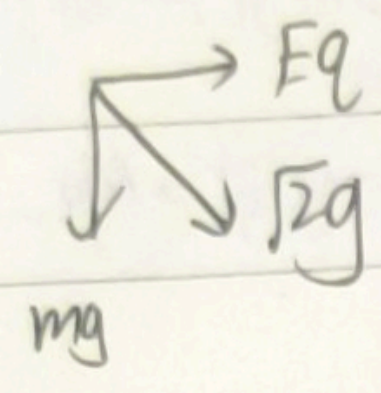
$$\begin{cases} mg + qE + F = \frac{mv^2}{L} \\ T \geq 0 \end{cases}$$

Date. $G' = 3mg$
 $g' = 1.5g$
 $V \geq \sqrt{3gL}$
 $V \geq \sqrt{1.5gL}$



$$\sqrt{2}mg + F = \frac{mv^2}{L} \quad F \geq 0$$

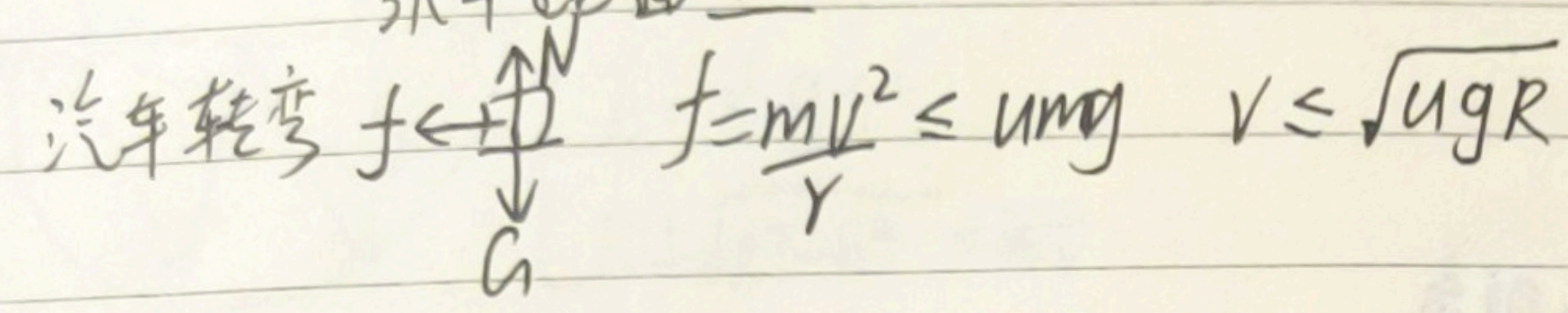
$$V \geq \sqrt{\sqrt{2}gL}$$



$$mg(l + \frac{\sqrt{2}}{2}l) + qE \frac{\sqrt{2}}{2}l = \frac{1}{2}m(v_0^2 - v^2)$$

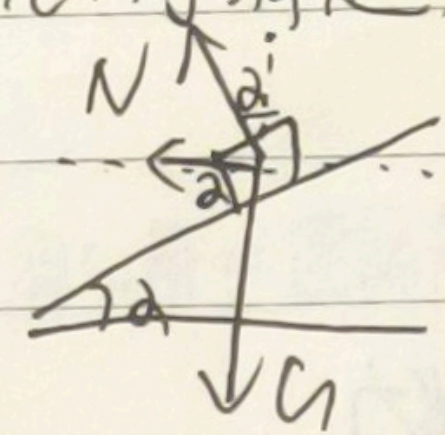
水平面圆周运动 必有侧向滑动趋势

水平路面上



$$f = \frac{mv^2}{R} \leq \mu mg \quad v \leq \sqrt{\mu g R}$$

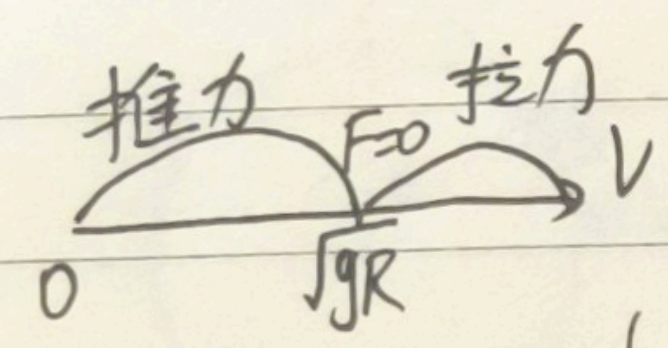
倾斜路面上 无侧滑趋势 v_0



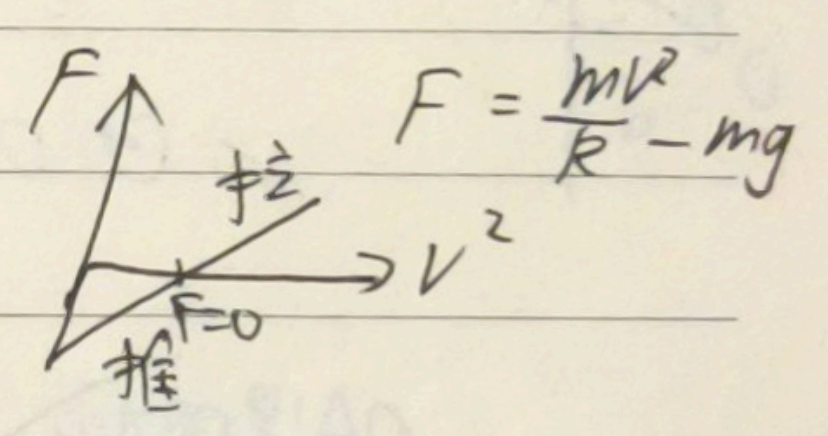
$$\begin{aligned} N \sin \alpha &= \frac{mv_0^2}{R} \\ N \cos \alpha &= mg \end{aligned}$$

$$v_0 = \sqrt{gR \tan \alpha} \quad \text{与 } \mu \text{ 无关}$$

杆模型



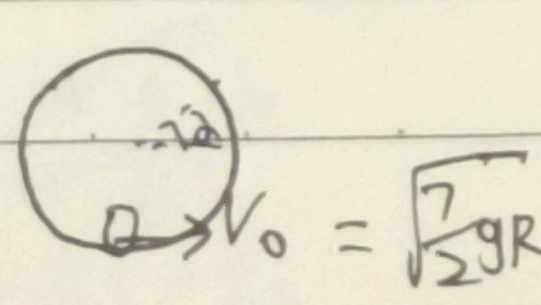
以向下为正 $mg + F = \frac{mv^2}{R}$ F 向下
 $mg - (-F) = \frac{mv^2}{R}$ F 向上



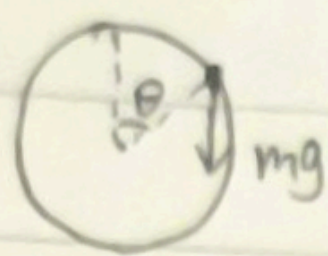
$$\begin{cases} 0 \leq v < \sqrt{gR} & \text{推力 } v \uparrow \text{ 推力 } \downarrow \\ v > \sqrt{gR} & v \uparrow F \text{ 拉 } \downarrow \end{cases}$$

绳不松了也 $N=0$ 临界

$$\begin{aligned} v_0 &\leq \sqrt{2gL} \\ v_0 &\geq \sqrt{5gL} \end{aligned}$$



$$\begin{aligned} mg \sin \alpha &= \frac{mv^2}{R} \\ -mg(l + \sin \alpha) &= \frac{1}{2}m(gR \sin \alpha - \frac{7}{2}gR) \\ \Rightarrow \alpha &= 30^\circ \end{aligned}$$



$$T + mg \cos \theta = \frac{mv^2}{R}$$

$$mg(1 - \cos \theta) = \frac{1}{2}mv^2 - \frac{1}{2}m(gR)$$

$$T = 3mg(1 - \cos \theta)$$

$$\text{对人 } N = Mg - T \cos \theta = Mg - 3mg(1 - \cos \theta) \cos \theta$$

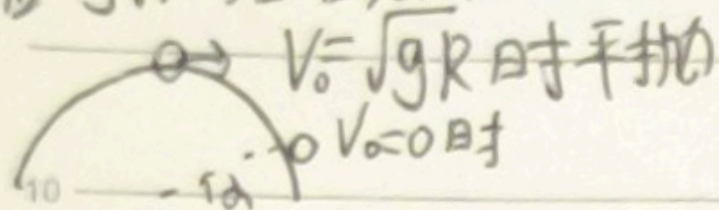
$$\geq Mg - 3mg \frac{1}{4} = Mg - \frac{3}{4}mg$$

$$T_{\max} - mg = \frac{mv_{\max}^2}{R} \quad v_{\max} = \sqrt{5gR}$$

$$\text{对人 } N = Mg + \frac{mv^2}{R} = Mg + 5mg$$

$$\text{故人对地压力 } N \in [Mg - \frac{3}{4}mg, Mg + 5mg]$$

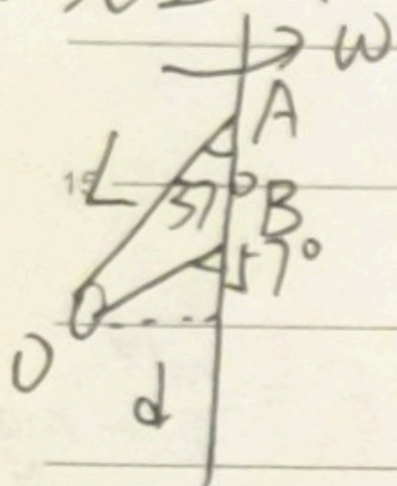
是否可以沿圆弧到 N 点?



$$mg \sin \alpha = \frac{mv^2}{R}$$

$$mg(1 - \sin \alpha) = \frac{1}{2}mv^2 \quad \sin \alpha = \frac{2}{3} \text{ 时脱离}$$

双绳竖直平面内做水平圆周运动



① OA 绷紧 OB 恰好松弛无张力

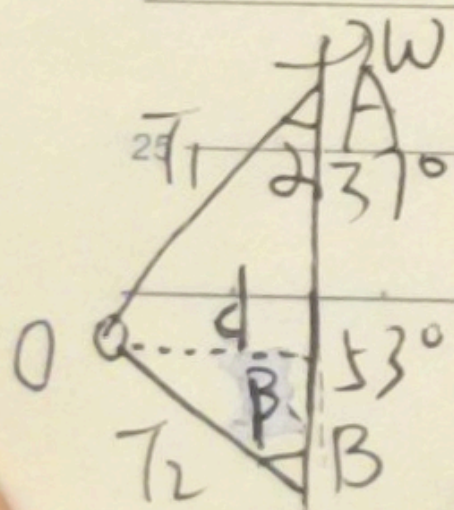
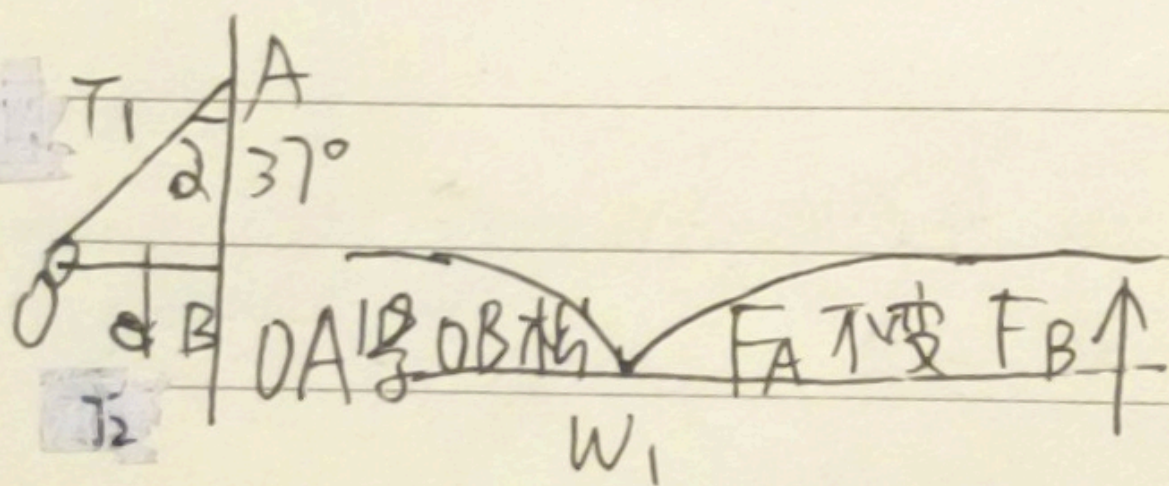
$$mg \tan 37^\circ = m\omega_1^2 d \quad d = L \sin 37^\circ \quad \omega_1 = \sqrt{\frac{3g}{4L}}$$

② OB 绷紧 OA 恰好伸直无张力

$$mg \tan 57^\circ = m\omega_2^2 d \quad \omega_2 = \sqrt{\frac{20g}{9L}}$$

OA 紧 OB 松 ω_1 ω_2 $\omega_1 < \omega < \omega_2$ 都紧 $\omega > \omega_2$ OA 松

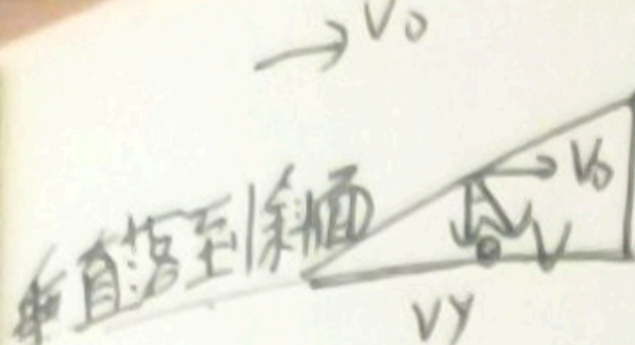
$$mg \tan \alpha = m\omega_1^2 d \quad \omega_1 = \sqrt{\frac{3g}{4d}}$$



$$T_1 \sin \alpha + T_2 \sin \beta = m\omega^2 d$$

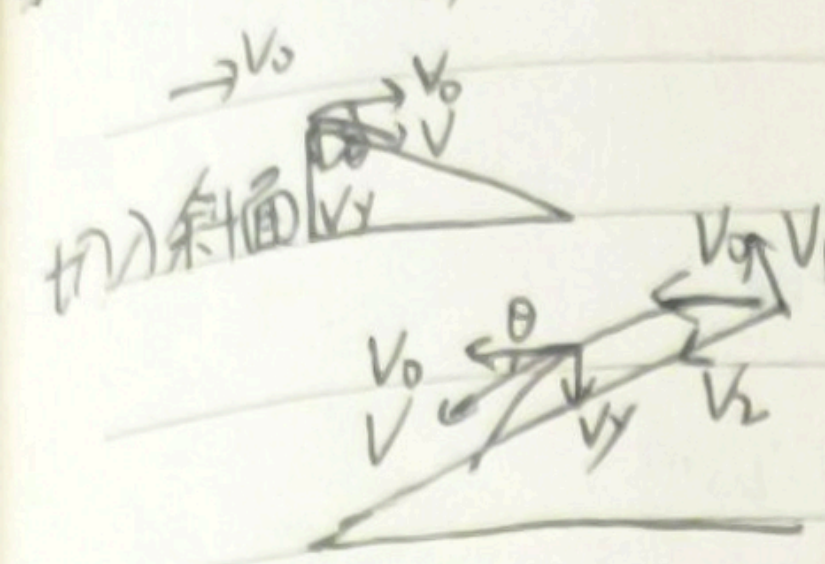
$$T_1 \cos \alpha = T_2 \cos \beta + mg$$

$$\omega \uparrow \quad T_1 \uparrow \quad T_2 \uparrow$$



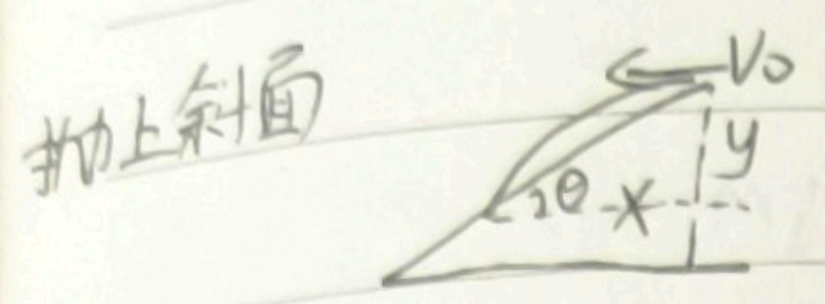
$$v_x = v_0 \quad v_y = gt \quad v = \sqrt{v_x^2 + v_y^2} \quad \tan \theta = \frac{v_y}{v_x} = \frac{v_0}{gt} \quad t = \frac{v_0}{g \tan \theta}$$

Date. NO.



$$t = \frac{v_1}{a}, \quad v_1^2 = 2ad \quad t = \frac{v_0 \tan \theta}{g} \quad d = \frac{v_0^2 \sin \theta \tan \theta}{2g}$$

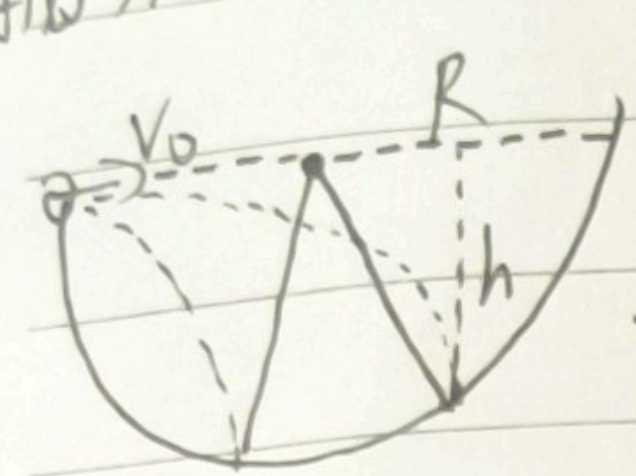
$$v_y = gt = \frac{v_0 \tan \theta}{g}$$



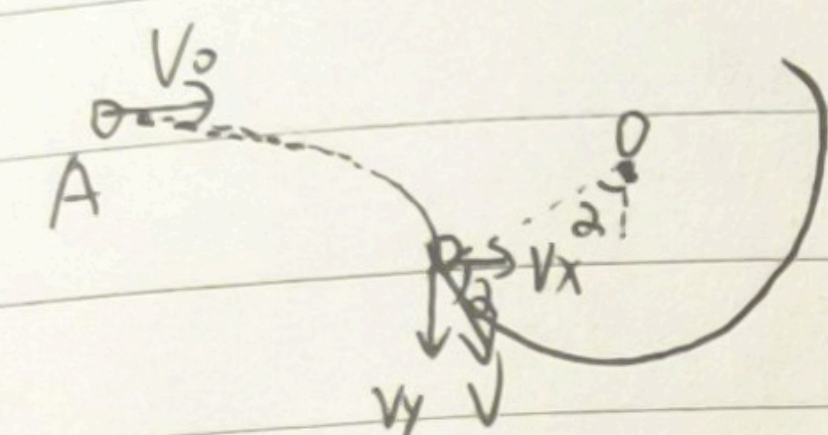
$$\begin{cases} x = v_0 t \\ y = \frac{1}{2} g t^2 \\ x^2 = x^2 + y^2 \end{cases} \quad \tan \theta = \frac{y}{x} = \frac{gt}{2v_0}$$

$$t = \frac{2v_0 \tan \theta}{g}$$

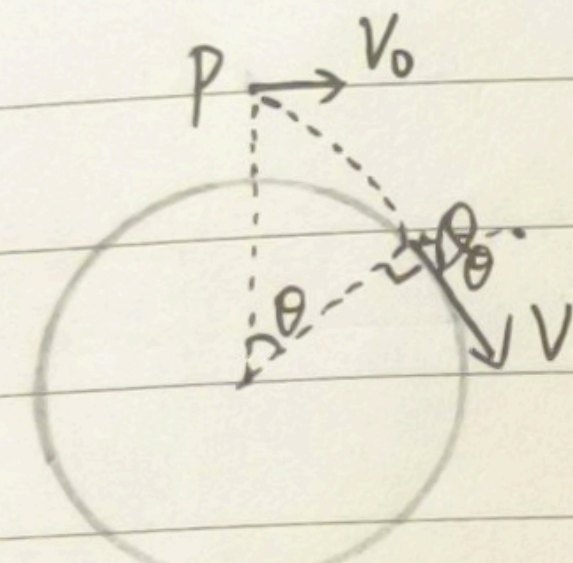
抛入圆3A



$$\begin{cases} h = \frac{1}{2} g t^2 \\ R \pm \sqrt{R^2 - h^2} = v_0 t \end{cases}$$



圆心角 = 速偏角



三类光滑斜面下滑时间比较

等高斜面

同底斜面

同圆周的斜面 一等时圆

$$L = \frac{1}{2} a t^2 \quad a = g \sin \theta \quad L = \frac{h}{\sin \theta}$$

$$L = \frac{1}{2} a t^2 \quad a = g \sin \theta \quad t = \frac{d}{\cos \theta}$$

$$\sin 2R = \frac{1}{2} g \sin \theta t^2 \quad t_1 = t_2 = t_3$$

$$t = \frac{1}{\sin \theta} \sqrt{\frac{2h}{g}}$$

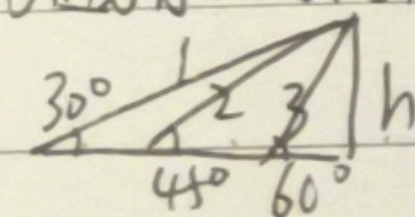
$$t = \sqrt{\frac{4d}{g \sin 2\theta}}$$

$$t_1 = t_2 = t_3$$

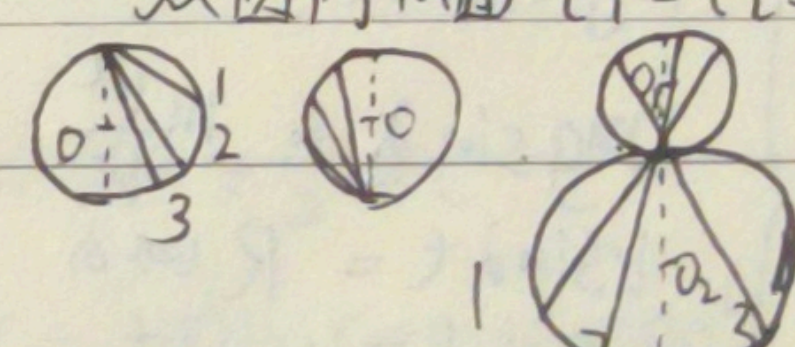
θ 越小 时间越长

θ = 45° 时 t min

双圆内斜面 t1 = t2 = t3

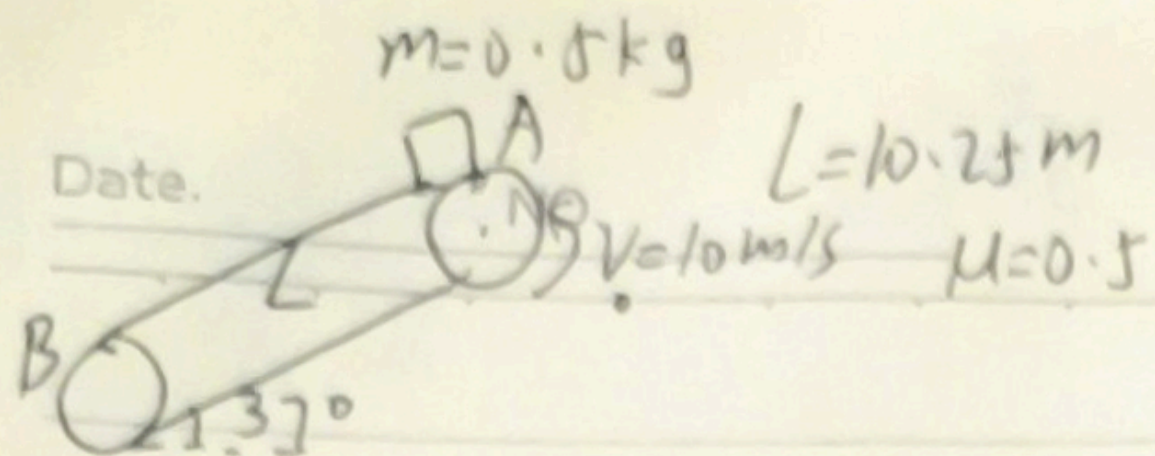


关于 45° 对称 t 相等



$$t_1 > t_2 > t_3$$

$$t_1 = t_3 > t_2$$



$$a_1 = 10\text{ m/s}^2 \quad a_2 = 2\text{ m/s}^2$$

$$g\sin\theta + \mu g\cos\theta \quad g\sin\theta - \mu g\cos\theta$$

$$t_1 = \frac{V_0}{a_1} = 1\text{ s} \quad S_1 = \frac{1}{2} V_0 t_1 = 5\text{ m}$$

$$L - S_1 = V_0 t_2 + \frac{1}{2} a_2 t_2^2 \quad t_2 = 0.5\text{ s}$$

$$\Delta X_1 = V_0 t_1 - X_1 = 5\text{ m}$$

$$\Delta X_2 = X_2 - V_0 t_2 = 0.25\text{ m}$$

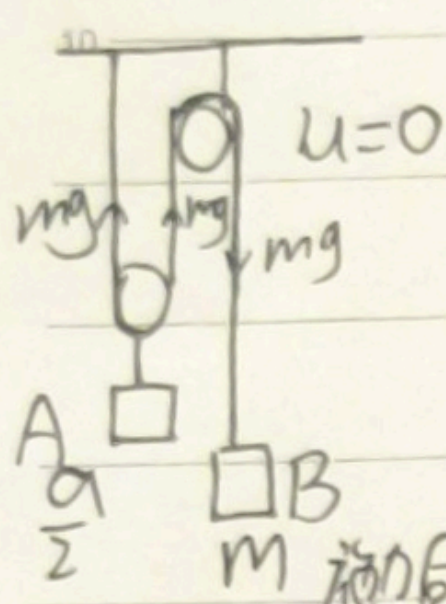
$$\Delta X_1 > \Delta X_2$$

故痕迹长度为 5m

$$W_{\text{带对块}} = f X_1 - f X_2 = \Delta E_{\text{机}} = \Delta E_k + \Delta E_p$$

$$W_{\text{电机对带}} = f V_0 t_1 - f V_0 t_2 = 10\text{ J} = \Delta E_{\text{电}} = \Delta E + Q$$

$$Q = f \times \overset{L-S_1}{5.25}\text{ m} = 10.5\text{ J}$$



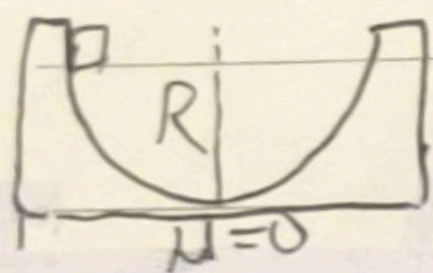
$u=0$ 开始时静止, 现对 B 施加一向下大小为 mg 的恒力 F

A 和滑轮总重 $2m$

$$\text{施力后, 对 B 由 } F + mg - T = ma$$

$$\text{对 A 及动滑轮 } 2T - 2mg = 2m \frac{a}{2}$$

$$\text{解得, } a = \frac{2}{3}g \quad T = \frac{4}{3}mg$$

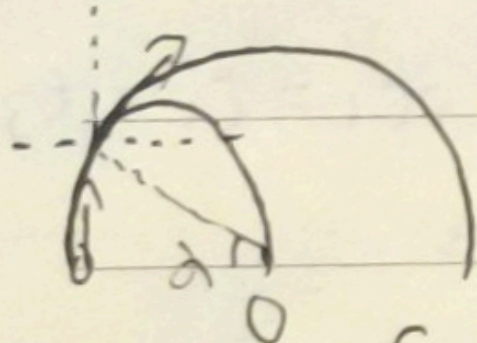


$$mgR = \frac{1}{2} m V_1^2 + \frac{1}{2} M V_2^2$$

$$mV = MV_2$$

$$N - mg = \frac{m(V_1 + V_2)^2}{R}$$

$$N = mg + \frac{m}{R} \left[\sqrt{\frac{2gR}{m+M}} \left(\sqrt{m} + \sqrt{\frac{m}{M}} \right) \right]^2$$



$$-mgR \sin \alpha = \frac{1}{2} m (v^2 - v_0^2)$$

$$mg \sin \alpha = \frac{m v^2}{R}$$

$$v \sin \alpha t = R \cos \alpha$$

$$v \cos \alpha = v \cos \alpha t - \frac{1}{2} g t^2$$

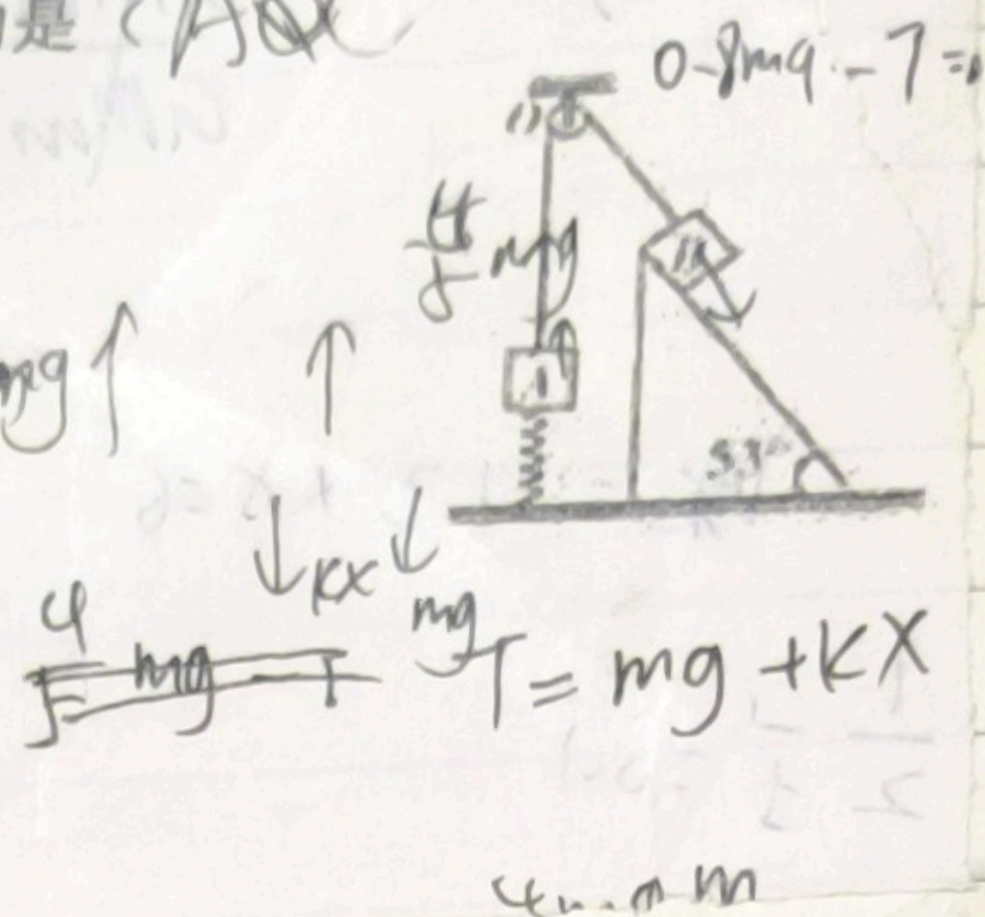
14. 如图所示, 劲度系数为 k 的竖直轻弹簧的下端固定在水平面上, 上端与物块 A 连接, 物块 B 与物块 A 之间用一绕过定滑轮 O 的轻绳连接, B 放在固定斜面的上端 (用外力控制 B)。OA 绳竖直, OB 绳与倾角为 53° 的足够长的固定斜面平行, 且轻绳恰好拉直而无作用力。A、B 的质量均为 m , A、B 均视为质点, 重力加速度大小为 g , 取 $\sin 53^\circ = 0.8$, $\cos 53^\circ = 0.6$, 不计一切摩擦。现将 B 由静止释放, B 沿斜面下滑, 弹簧始终处在弹性限度内。下列说法正确的是 (A) ~~(A)~~

~~A.~~ 释放 B 后的瞬间, 轻绳的弹力大小为 $\frac{2}{5}mg$

B. 当 A 的速度最大时, 弹簧处于拉伸状态

C. A 的最大速度为 $\sqrt{\frac{8mg^2}{25k}}$

D. 当 B 处于最低点时, 弹簧处于压缩状态



A. 释放前对 A $kx_1 = mg$

释放后 对 A、B $kx_1 - mg + mgsin 53^\circ = 2ma$ $a = \frac{2}{5}g$

对 A $T_1 + kx_1 - mg = ma$ $T_1 = \frac{2}{5}mg$

B. V_{Amax} 时 $a_A = a_B = 0$ $T_2 = mgsin 53^\circ < mg$ 故弹簧压缩

C. V_{Amax} 时 对 A 由平衡条件 $T_2 + kx_2 = mg$ $x_2 = \frac{mg}{5k}$

由动能定理

$$\frac{k}{2}(x_1^2 - x_2^2) + mg(x_1 - x_2)sin 53^\circ - mg(x_1 - x_2) = \frac{1}{2}2mV_m^2 - 0$$

$$V_m = \sqrt{\frac{8mg^2}{25k}}$$

D. 设当 B 处于最低点时压缩 x_3

$$\frac{k(x_1 + x_3)}{2}(x_1 - x_3) + mg(x_1 - x_3)sin 53^\circ - mg(x_1 - x_3) = 0$$

$$x_3 = -\frac{3mg}{5k} < 0 \text{ 不成立}$$

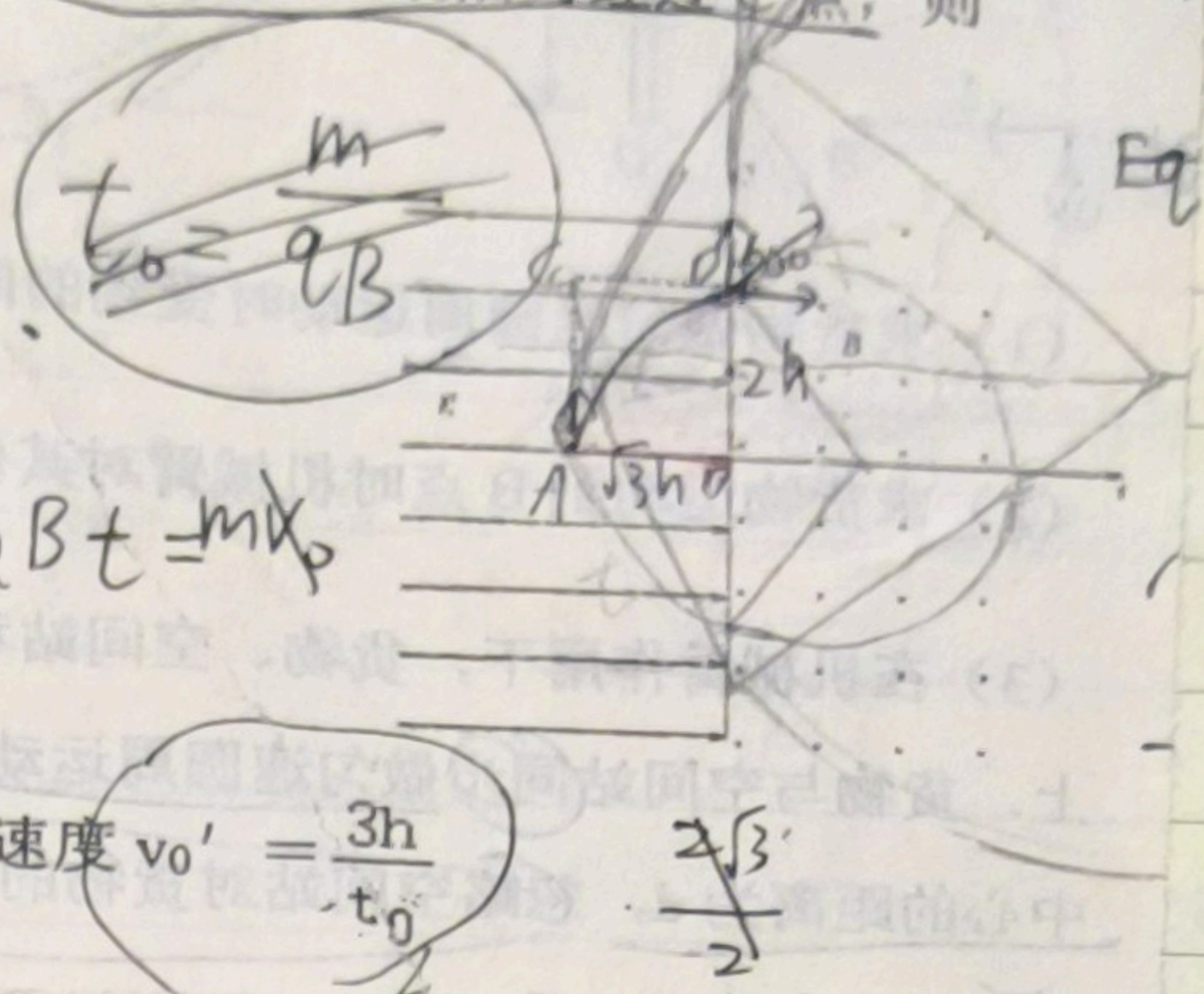
故拉伸。

13. 空间存在平面直角坐标系 xOy , 在 $x < 0$ 区域内有沿 x 轴正方向的匀强电场, 在 $x > 0$ 区域内有垂直平面向外的匀强磁场, 在第二象限内有矩形 $OACD$, $OA = \sqrt{3}h$, $OD = 2h$. 质量为 m , 电量为 q 的带正电粒子 (不计重力) 从 A 点沿 y 轴正方向以某速度射入第三象限, 经 t_0 时间后由 D 点进入磁场, 又经一段时间射出磁场后又回到 A 点. 现只改变粒子自 A 点出射速度大小至 v , 粒子经过一段时间运动后可经过 C 点, 则

A. 匀强电场的场强大小为 $\frac{2\sqrt{3}mh}{qt_0^2}$

B. 匀强磁场的磁感应强度大小为 $\frac{\sqrt{3}m}{3qt_0}$

C. 能使粒子以最短时间从 A 点运动至 C 点的初速度 $v_0' = \frac{3h}{t_0}$



$$\sqrt{3}h = \frac{1}{2}at_0^2 \quad a = \frac{qE}{m}$$

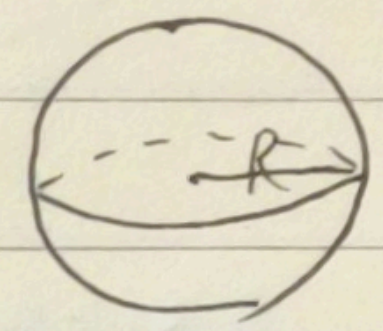
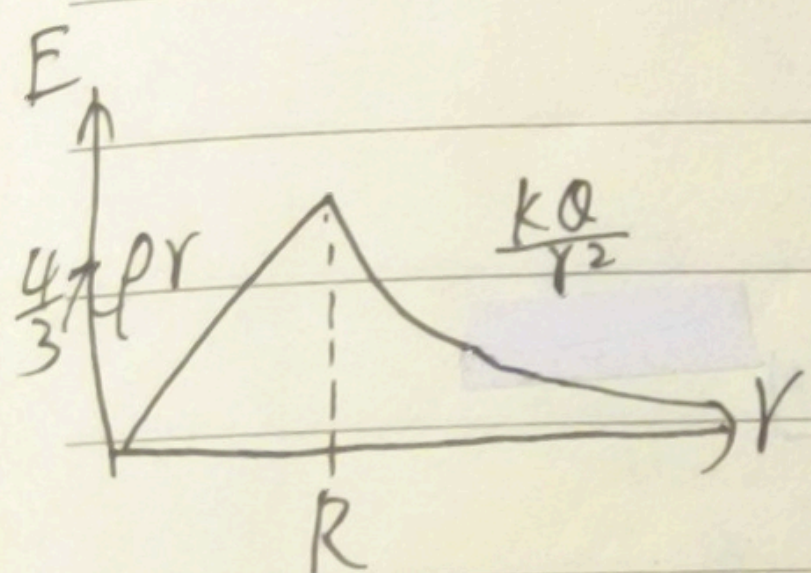
$$E = \frac{2\sqrt{3}mh}{qt_0^2}$$

$$R = \frac{2h}{\sin\theta} = \frac{mv}{qB} \quad v = \frac{V_0}{\cos\theta} = \frac{4h}{t_0} \quad \therefore B = \frac{\sqrt{3}m}{qt_0}$$

$$V_x = \frac{2\sqrt{3}h}{t_0} \quad \sin\theta = \frac{V_x}{V'} = \frac{2\sqrt{3}h}{V't_0} \quad qV'B = \frac{mV'^2}{R} \quad R' = \frac{mV'}{qB} \quad R'\sin\theta = 2h$$

由对称性 $y = 2R'\sin\theta = 4h \quad \therefore y_1 = 3h \quad y_2 = -h \quad \therefore V_0' = \frac{y_1}{t_0} = \frac{3h}{t_0}$

正则动量



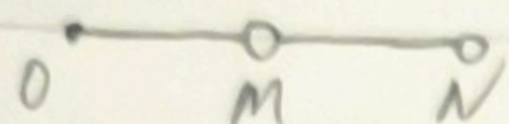
电球场强

等效绳长长 $r_0 \omega^2 = \frac{g}{4\omega^2}$

张力的平行四边形定则时不在平行四边形中是理想的
太空中圆周围圆 v 越大油水越易分离

Date.
 R越小电容充放电越快

平抛在空间中对称线处最远



以M为圆心做圆周运动

5
图象法一面积 $\left\{ \begin{array}{l} F-x \\ a-x \\ v-x \\ p-v \end{array} \right.$

动能定理 多过程曲线运动

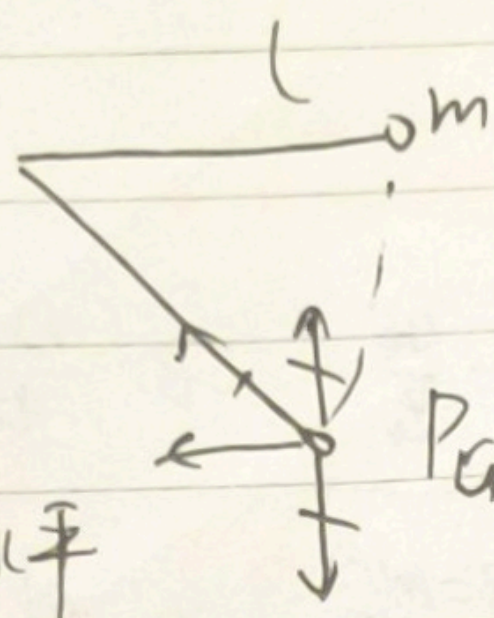
10
P定义式 $P = \frac{W}{t}$

$$P = F \cdot v \cos \alpha$$

$$\bar{P} = F \bar{v}$$

$$P_{\text{瞬}} = F v_{\text{瞬}}$$

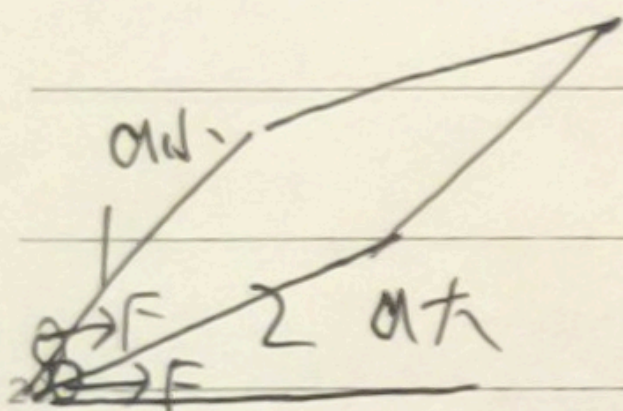
E-t图斜率为对应功率



$\left\{ \begin{array}{ll} E_k-t & P_{\text{外力}} \\ E_{p_h}-t & P_G \\ E-t & P_{\text{其他}} \text{ 如 } P_f \text{ 等} \end{array} \right.$

15
极限法 $v = \sqrt{\frac{2MgR}{M+R}} \quad M \gg m \rightarrow v = \sqrt{2gR}$

特殊值法 有k, n, 类参数、比值、直接赋值。



$$N = mg \cos \alpha + F \sin \alpha$$

$$f = \mu mg \cos \alpha + \mu F \sin \alpha$$

$$W_f = -f s = -\mu mg s \cos \alpha - \mu F s \sin \alpha = -\mu mg x - \mu F y$$

$$\Delta E_{k1} = \Delta E_{k2}$$

$$t_1 > t_2$$

